Spatiotemporal Enhancement and Interlevel Fusion Network for Remote Sensing Images Change Detection

Yanyuan Huang, Xinghua Li, *Senior Member, IEEE*, Zhengshun Du, and Huanfeng Shen, *Senior Member, IEEE*

***Abstract*— Remote sensing (RS) image change detection (CD) plays a crucial role in monitoring surface dynamics; however, current deep learning (DL)-based CD methods still suffer from pseudo changes and scale variations due to inadequate exploration of temporal differences and under-use of multiscale features. Based on the aforementioned considerations, a spatiotemporal enhancement and interlevel fusion network (SEIFNet) is proposed to improve the ability of feature representation for changing objects. First, the multilevel feature maps are acquired from the Siamese hierarchical backbone. To highlight the disparity in the same location at different times, the spatiotemporal difference enhancement modules (ST-DEMs) are introduced to capture global and local information from bitemporal feature maps at each level. Coordinate attention (CA) and cascaded convolutions are adopted in subtraction and connection branches, respectively. Then, an adaptive context fusion module (ACFM) is designed to integrate interlevel features under the guidance of different semantic information, constituting a progressive decoder. Additionally, a plain refinement module and a concise summation-based prediction head are employed to enhance the boundary details and internal integrity of CD results. The experimental results validate the superiority of our lightweight network over eight state-of-the-art (SOTA) methods on LEVIR-CD, SYSU-CD, and WHU-CD datasets, both in accuracy and efficiency. Also, the effects of different types of backbones and differential enhancement modules are discussed in the ablation experiments in detail. The code will be available at https://github.com/lixinghua5540/SEIFNet.**

***Index Terms*— Change detection (CD), deep learning (DL), difference enhancement, feature fusion, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) aims to identify object variations within the same area over different time periods. CD from remote sensing (RS) images, widely employed in environment monitoring [1], resource investigation [2], urban expansion analysis [3], and disaster assessment [4], face challenges. Diverse sensors introduce discrepancies in imaging conditions, including time and angle differences, resulting in pseudo changes like seasonal variations and shadows; moreover, higher resolution images often contain complex objects, which pose greater challenges for CD.

The conventional methodologies have established the foundation for CD, including algebra-based, transformation-based, and classification-based approaches. The algebra-based methods involve arithmetic operations such as image differencing [5], image ratio [6], and image quantification [7]. They calculate the pixel differences between bitemporal images to generate a difference map of changed and unchanged pixels by thresholding or cluster analysis. Transformation-based methods, such as principal component analysis (PCA) [8] and tasseled cap transformation (KT) [9], transform features into specific spaces to highlight the changed pixels. Classification-based methods, employing techniques like support vector machine (SVM) [10] and random forest (RF) [11], discern changed pixels by comparing classification maps between bitemporal images; therefore, the accuracy of CD by classification-based methods heavily relies on classification. Although the traditional methods are straightforward and unsupervised, they often rely on empirical feature selection and manual threshold adjustments. Their performance often lags due to the intricate object distribution in high-resolution images. Albeit machine learning (ML)-based methods have made advancements in CD from high-resolution RS images [12], [13], they still face limitations in modeling complicated features.

Deep learning (DL) has revolutionized RSCD by leveraging the capability of neural networks to use deep-level features. Indeed, it enables end-to-end CD and surpasses traditional and ML-based methods for high-resolution RS images. Various networks have been employed in CD, such as convolutional neural networks (CNNs) [14], generative adversarial networks (GANs) [15], recurrent neural networks (RNNs) [16], and then transformers [17]. Among them, CNNs have been dominant in the field of CD, owing to their outstanding performance. The earlier DL-based methods adopt a unified input by connecting coregistered bitemporal images to the network. For example, the bitemporal images are connected in the channel dimension as input [18]. It is known as early fusion

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(EF) networks, which suffer from the inadequate extraction of respective single-temporal features. Later fusion (LF) networks, in contrast to EF networks, employ parallel branches with shared weights to extract features separately from single-temporal images. Numerous studies have consistently shown that the LF networks outperform the EF networks in terms of the specificity for CD [19], [20], [21], [22]. Yet, a major drawback is the potential information loss, limiting suitability for complex, high-resolution RS images. Researchers have, therefore, attempted to improve the extraction and use of deep features [23], [24]. For example, SNUNet [25] designed dense skip connections to combine multilevel features, which mitigated the loss of localization information in deep layers. As the network deepened, both the parameter counts and complexity, however, experienced a substantial surge. The convolutional computation has been optimized to reduce complexity, e.g., depth-separated convolutional [26]. Another common idea for improvement is to increase the receptive field of convolution, such as the introduction of pyramid structures [27], [28], dilated convolution [29], [30], and dynamic convolution [31], which achieve a similar effect to deepen the network.

While DL-based methods have achieved significant advancements, it is important to acknowledge the existing limitations. On the one hand, it is the inadequate use of temporal information. The fundamental objective of CD is to detect and characterize differences. Existing methods, however, often focus on feature extraction, overlooking differences between bitemporal images. On the other hand, multiscale feature enhancement should be further improved. Numerous prevailing approaches to multiscale feature fusion heavily depend on the specific structure of modules while disregarding interscale feature associations. Generally, the low-level features aid target localization, and the high-level features provide semantic context. Recent studies [32] have demonstrated that the complementarity of high-low-level communication through the interaction of features can enhance the detailed characterization of CD results. This is a valid idea for detecting scale variations in high-resolution RS images.

To address the aforementioned challenges, a novel spatiotemporal enhancement and interlevel fusion network (SEIFNet) is proposed. First, the bitemporal images are fed into the backbone to obtain multiscale feature maps. Then, to highlight the changed regions, a spatiotemporal difference enhancement module (ST-DEM) with a dual branch structure is designed to learn target-level global changing information and fine-grained local content between bitemporal feature maps at the same scale. Then, the adaptive context fusion module (ACFM) and refinement module (RM) are adopted for progressive decoding. Finally, a summation-based prediction head is employed to aggregate feature maps from different scales, facilitating the CD results of bitemporal images. The main contributions of our work can be summarized as follows:

1) A novel SEIFNet based on the Siamese network is developed for pixel-level RSCD. The encoder focuses on spatiotemporal characteristics to enhance the differences between bitemporal images. The decoder realizes the complementary advantages of multiscale features by interlevel fusion. Besides, a class balance loss function is designed to tackle class imbalance in CD tasks.
2) ST-DEM employs a dual-branch structure to learn global and local information on bitemporal features separately. Coordinate attention (CA) is used in the subtraction branch to capture long-term dependencies of differences. The spatial details are extracted with cascaded convolution in the connection branch. The module is designed to highlight changed regions and suppress noise interference.
3) A progressive decoder, consisting of ACFM and RM, is devised to merge and refine multiscale features. The ACFMs achieve adaptive feature fusion through the reciprocal guidance of diverse contextual information embedded in interlevel features, while the rms further optimizes boundary details. Such ways enable a better reconstruction of the detailed information of objects.

The rest is organized as follows. In Section II, a brief review of related work is presented. The framework and details of our proposed network are described in Section III. Then, the experimental analyses are presented in Section IV. In Section V, the effectiveness and limitations of the methodology are discussed. Finally, it is concluded in Section VI.

II. Related Work

In recent DL-based RSCD studies, the introduction of an attention mechanism and a multiscale feature fusion strategy are two popular approaches. In this section, the two techniques are briefly reviewed in the application of the CD task.

A. Attention Mechanism in CD

The attention mechanism has been extensively employed in CD to guide the model in focusing on the regions of interest that exhibit changes [33]. For instance, the image fusion network (IFN) [34] incorporates both channel attention and spatial attention in feature fusion to improve boundary completeness and internal compactness. Similarly, Shi et al. [35] introduced the convolutional block attention module (CBAM), a combination of channel attention and spatial attention, in the metric module to obtain more discriminative features.

Many existing studies on the attention mechanism, however, primarily address long-range dependencies in the channel and spatial domains but often overlook the crucially temporal relationship for CD. The essence of CD lies in observing differences at the same geographical position across different times. The feasibility of generating discriminative features from temporal information has been demonstrated. For example, the JoinAtt module in the dual-branch multilevel intertemporal network (DMINet) [36] skillfully realized the feature interaction, steering attention to the genuine changes. Chen and Shi [37] proposed a pyramid spatial-temporal attention module (PAM) with the self-attention mechanism. It captures multiscale features and finer details of change-related objects. Likewise, the efficient spatial-temporal attention module (ESTAM) in the densely attentive refinement network (DARNet) [38] used self-attention to extract spatiotemporal context information from bitemporal images.

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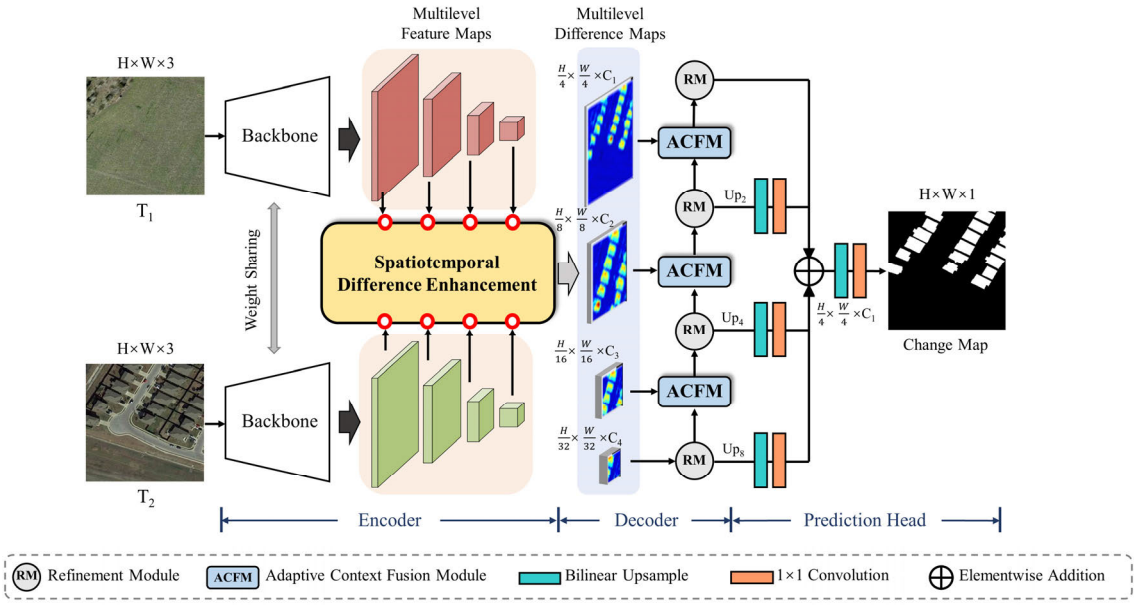


Fig. 1. Architecture of SEIFNet. The network comprises three components: hierarchical spatiotemporal enhanced encoder, progressive decoder, and summation-based prediction head.

To reduce the cost, crisscross attention is incorporated to model long-range relationships. But even so, self-attention consumes significant GPU memory due to the consideration of associations among all pixels [39]. Thus, it is necessary to effectively capture long-range dependencies spanning channel, spatial, and temporal dimensions.

B. Multiscale Feature Fusion Strategy in CD

Multiscale feature fusion refers to the aggregation of feature maps at different levels. During the feature learning process, low-level features, such as edges and corners, provide fine-grained details, while high-level features capture more global information. A multiscale feature fusion strategy is, therefore, employed to amalgamate the strengths of deep and shallow features across various levels.

One of the most straightforward approaches is to sample and normalize features from each layer to the same scale and then combine them into a unified representation, like ChangeFormer [40]; however, excessive sampling intervals are detrimental to the restoration of feature details. To address this issue, Li et al. [41] proposed a temporal feature interaction and guided refinement network (TFI-GR) to aggregate temporal change information from multilevel features, generating change maps in a coarse-to-fine fuse procedure. Beyond that, both the multiscale feature progressive fusion network (MFPF-Net) [42] and the multiscale diff-changed feature fusion network (MSDFFN) [43] used a reweighting strategy. This approach enhances feature representation by selecting informative features from diverse scales for fusion; however, the aforementioned methods directly fuse features from all scales, presenting a challenge of cross-scale semantic disparity. Consequently, there is a growing exploration of feature fusion between adjacent scales. For instance, a masked aggregation module in the intrascale cross interaction and interscale feature fusion network (ICIF-Net) [44] was proposed to highlight the respective properties of interest in reconstructed features for interscale feature fusion. Besides, A2Net [45] introduced a neighbor aggregation module to enhance the representation capability of extracted temporal features from low-mid-high levels. The above researches indicate that the fusion of multiscale feature maps can be profitable for comprehensive details and localizations.

III. Methodology

In this section, first, the overall framework of SEIFNet is described. It is an encoder-decoder structure with a summation-based prediction head for RSCD. Then, ST-DEM and ACFM are presented in detail. Finally, the loss function designed for mitigating class imbalance is introduced.

A. Architecture Overview

The overall architecture of SEIFNet is illustrated in Fig. 1. The proposed SEIFNet offers a flexible and efficient approach for CD by enhancing the spatiotemporal differences between bitemporal high-resolution RS images, and it provides a new idea for interlevel feature fusion and refinement. In the encoder part, the multiscale features of input images are first extracted using a Siamese backbone with two parallel branches. The pretrained ResNet18 [46] is chosen as the backbone of SEIFNet.

The backbone generates multilevel feature maps at four stages with sizes of 64×64, 32×32, 16×16, and 8×8. After that, ST-DEM processes multilevel feature pairs from the backbone separately. This stage is achieved by augmenting the spatial differences between the bitemporal features, thus highlighting the changed regions and suppressing the interference. Next, we employ a hierarchical feature decoding approach where interlevel feature maps are merged with ACFM for contextual information fusion. The fused feature is then refined

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with a refinement module consisting of CBAM [47] and convolutional operations. This process is repeated to obtain multiscale refined features at each level. Finally, the outputs of the refinement modules are resampled to the same scale and summed together.

Overall, SEIFNet focuses on changes from the perspective of spatiotemporal differences and minimizes information loss through multiscale feature fusion. Both are critical for CD tasks.

B. Spatiotemporal Difference Enhancement Module (ST-DEM)

Currently, numerous methods exist for generating difference maps from bitemporal images, including direct subtraction, summation, concatenation, as well as enhanced mechanisms. Every method, however, has its inherent limitations. Subtraction-based methods tend to introduce large numbers of noise and can potentially confuse subtle change features, leading to missed detections and false positives. Connection-based approaches retain information on channel dimensions, but they lack interpretability and highly rely on the labeled datasets. Toward this end, ST-DEM, with a dual-branch structure, is designed to learn target-level global change-related information and fine-grained local context between bitemporal feature maps.

ST-DEM consists of a subtraction branch and a connection branch, as shown in Fig. 2. The subtraction branch computes the absolute difference (DI^i) between bitemporal feature maps at the same scale as its input, while the connection branch combines features (F_1^i and F_2^i) in the channel dimension for subsequent progress. The former uses CA to leverage the spatiotemporal difference between bitemporal feature maps. In the latter branch, two convolutional blocks are used to detect local contextual information for supplementation and reduce the noise of interference. The whole implementation can be represented as

$$\mathrm{DI}_i = \left|F_1^i - F_2^i\right| \tag{1}$$

$$F_D^i = \mathrm{CA}(\mathrm{DI}_i) \tag{2}$$

$$F_C^i = \mathrm{Conv1}\left(\mathrm{Conv3}\left(\left[F_i^1, F_i^2\right]\right)\right) \tag{3}$$

$$D^i = F_D^i + F_C^i \tag{4}$$

where, $|\cdot|$ and $[\cdot]$ represents the absolute value of subtraction and the connection of channel dimension, respectively. CA is coordinate attention, Conv1 and Conv3 are convolutional operations with kernel sizes of 1 and 3, respectively, followed by batch normalization and ReLU. After that, F_D^i and F_C^i are combined through element-wise addition to obtain the enhanced difference feature (D^i), which incorporates both long-range spatiotemporal information and local contextual information.

In the subtraction branch, an attention-based approach is adopted to capture global and local features. In the ordinary attention mechanism, global pooling and convolution are widely used; however, the pooling operation compresses features into one dimension directly, resulting in the loss of location-specific details, and the convolution is restricted

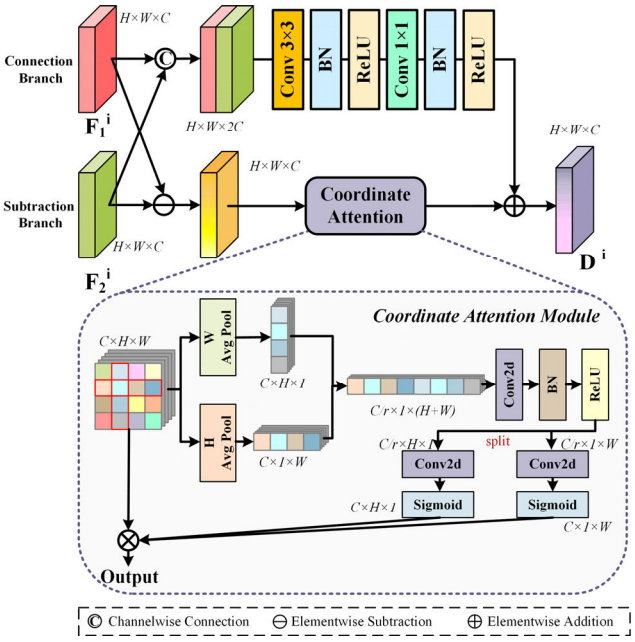


Fig. 2. Illustration of the ST-DEM.

by the receptive field, facing challenges in modeling long-range dependencies. To address the limitation of constrained extraction of global and local information, we adopt a novel attention mechanism introduced by Hou et al. [48] called *coordinate attention* (CA), which embeds position into channel attention. The CA consists of two steps: coordinate information embedding and CA generation, as shown at the bottom of Fig. 2. First, two pooling kernels with different spatial extents are applied to encode each channel horizontally and vertically. Second, the outputs of the pooling layers are concatenated and passed through a shared 1×1 convolution. The resulting tensor is then split into two separate tensors, which yield attention vectors with the same number of channels for the horizontal and vertical coordinates of the input X. The process can be written as follows:

$$z^h = \mathrm{GAP}^h(X), \quad z^w = \mathrm{GAP}^w(X) \tag{5}$$

$$f = \delta\left(\mathrm{Conv1}\left(\left[z^h; z^w\right]\right)\right) \tag{6}$$

$$f^h, f^w = \mathrm{Split}(f) \tag{7}$$

$$s^h = \sigma\left(\mathrm{Conv}_h^{1\times1}\left(f^h\right)\right), \quad s^w = \sigma\left(\mathrm{Conv}_w^{1\times1}\left(f^w\right)\right) \tag{8}$$

$$Y = X s^h s^w \tag{9}$$

where GAP^h and GAP^w denote the pooling functions for vertical and horizontal coordinates, respectively. δ is a nonlinear activation function, σ is the sigmoid function, $s^h \in R^{C\times1\times W}$ and $s^w \in R^{C\times H\times1}$ represent the corresponding attention weights, respectively.

We introduced CA to the CD task because of its ability to balance efficiency and spatial location awareness. The resulting feature maps are subsequently encoded into direction-aware and position-sensitive attention maps separately; thus, spatial differences between bitemporal feature maps can be accurately captured through CA.

The connection branch is composed of a 3×3 convolutional block, which is responsible for learning local information, and

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a 1 × 1 convolutional block, which reduces the number of channels to match the output of the subtraction branch. Each convolutional block is comprised of a convolutional operation followed by batch normalization and ReLU. It can be used to compensate for the missing details and noise interference that may be caused in the subtraction branch, reducing the dependence on a specific CD dataset. ST-DEM incorporates CA and successive convolutional blocks to capture both global and local information from bitemporal feature maps. The module effectively enhances the representation of change-related regions and suppresses irrelevant interferences. From that, four different scales of spatiotemporal enhanced difference maps are sequentially generated.

C. Adaptive Context Fusion Module (ACFM)

The refinement of multiscale targets in the decoder plays a crucial role, which is equal to the feature extraction process in the encoder. In conventional decoders, the aggregation of feature maps from different scales is typically performed through upsampling. It may, however, result in the loss of spatial details due to the significant information differences at different levels. To address this problem, ACFM is introduced to merge interlevel features, leveraging the contextual information from high-level features to guide the restoration of low-level features. The visual depiction of the progress is illustrated in Fig. 3(a).

ACFM is composed of three steps: feature upsampling, weight calculation, and feature reintegration. First, the high-level difference feature map d^{i+1} is upsampled with bilinear interpolation to match the resolution of the low-level difference feature map d^i. Additionally, a 1 × 1 convolutional layer with batch normalization and ReLU is employed to decrease the channel dimension of the upsampled feature map. As a result, a feature map, denoted as d_up^{i+1}, is obtained, which possesses size and dimensions identical to the low-level difference map. Then, the weights for integrating d^i and d_up^{i+1} are calculated by a specific pooling process, which can be described as follows:

$$w_h = \delta\Big(\text{FC}\Big(\text{MaxPool}\Big(d^i + d_{\text{up}}^{i+1}\Big)\Big) + \text{FC}\Big(\text{AvgPool}\Big(d^i + d_{\text{up}}^{i+1}\Big)\Big)\Big) \tag{10}$$

$$w_l = 1 - w_h \tag{11}$$

where MaxPool, AvgPool, FC, and δ represent the operation of the max pooling, average pooling, fully connected layer with two successive 1 × 1 convolution and sigmoid operation, respectively. w_h and w_l are the fusion weights of d^i and d_{up}^{i+1}, respectively. The original intention of such a setting is based on weight normalization, which aims to keep the data distribution balanced, thereby avoiding overfitting. Finally, the weights are multiplied by both the original low-level difference map and the upsampled high-level difference map, respectively. These weighted feature maps are then summed together to obtain the fused feature map (Fusion^i). The reintegration process involves combining the complementary information from different levels to enhance the overall representation.

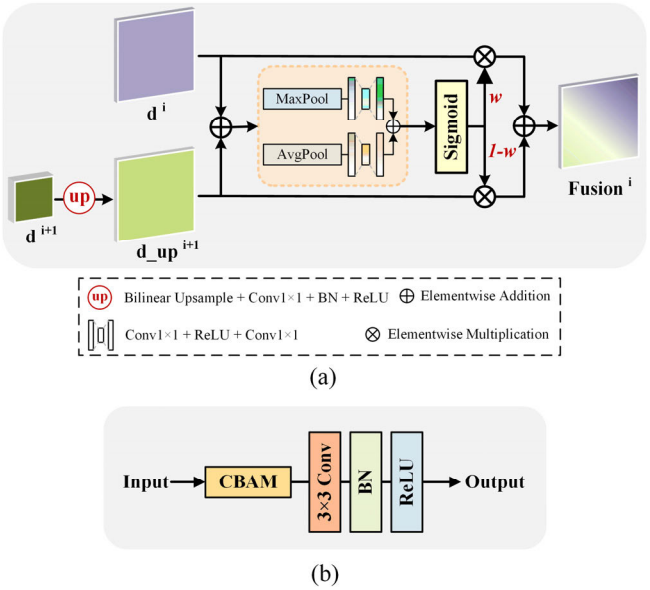


Fig. 3. Illustration of ACFM and RM. (a) ACFM. (b) RM.

Mathematically, it can be formulated as follows:

$$\text{Fusion}^i = w_h \times d^i + w_l \times d_\text{up}^{i+1}. \tag{12}$$

Based on the proposed ACFM, a coarse-to-fine multilevel feature fusion is implemented. To enhance the discrimination of change regions and suppress irrelevant changes, we use CBAM and a 3 × 3 convolution operation to guide the refinement of fused features, as shown in Fig. 3(b). This additional refinement module (RM) enhances the discrimination of the features and thus improves the CD results.

D. Class Balance Loss Function

In CD tasks, class imbalance is a common challenge, where the number of unchanged areas (negative samples) far exceeds the number of changed areas (positive samples) in the dataset. The imbalance can lead to the network focusing more on learning the abundant negative samples but neglecting the important positive samples that are relevant for CD. To mitigate the effects of class imbalance, a class balance loss function is used to place more emphasis on challenging scenes in the CD task. The loss function consists of two components: binary cross-entropy (BCE) loss [49] and dice coefficient (Dice) loss [50]. The BCE loss L_{BCE} and Dice loss L_{Dice} are formulated as

$$L_{\text{BCE}}(y, \hat{y}) = -\big(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\big) \tag{13}$$

$$L_{\text{Dice}}(y, \hat{y}) = 1 - \frac{2y\hat{y} + 1}{y + \hat{y} + 1} \tag{14}$$

where y and $\hat{y}$ are the ground true label and corresponding predicted change map, respectively. The value of pixels in y is 0 or 1, yet the value of pixels in $\hat{y}$ from 0 to 1, represents the probability of being predicted as changed or unchanged. The overall loss function is defined as

$$L(y, \hat{y}) = \lambda L_{\text{BCE}}(y, \hat{y}) + (1 - \lambda) L_{\text{Dice}}(y, \hat{y}) \tag{15}$$

where λ is a parameter that balances L_{BCE} and L_{Dice}. In the combined loss function, L_{Dice} focuses on handling class

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imbalance, and L_{BCE} is used to facilitate convergence. The appropriate weight parameter λ will be determined through subsequent experimental analysis.

IV. Experiments

This section describes the experiments and analysis in three parts. Three public CD datasets and implementation settings in our experiments are introduced in Sections IV-A and IV-B, respectively. In Section IV-C, our proposed method is compared with seven state-of-the-art (SOTA) methods, both quantitatively and qualitatively.

A. Datasets

Three representative CD datasets used in the experiments are LEarning, VIsion, and Remote sensing Change Detection (LEVIR-CD) [37], Sun Yat-sen University Dataset (SYSU-CD) [35], and WHU Building Dataset (WHU-CD) [51].

1) *LEVIR-CD:* This is a dataset for building CD, which contains 637 pairs of images with 1024 × 1024 spatial size and 0.5m spatial resolution. The original images were taken from 20 regions in Texas, USA, by using the Google Earth platform. Based on the official dataset split setting, each image was cropped into nonoverlapping patches with a size of 256 × 256 and obtained 7120/1024/2048 image pairs for training/validation/test, respectively.
2) *SYSU-CD:* This dataset includes various types of changes such as buildings, vegetation, roads, and sea constructions, which presents more challenges for the CD task. The dataset contains 20000 pairs of 0.5m aerial images with a size of 256 × 256 taken in Hong Kong from 2007 to 2014. For training/validation/testing sets, we divided patches into 12000/4000/4000 according to the ratio of 6:2:2.
3) *WHU-CD:* The public building CD dataset was generated from a pair of 0.7m aerial images with a size of 32507 × 15354. The original images were cropped into patches with a size of 256 × 256 and divided randomly into 6096/762/762 for training, validation, and testing.

B. Implementation Details and Metrics

1) Implementation Details: During our experiments, we trained the proposed model using the PyTorch framework on an NVIDIA Quadro RTX 5000 (16G) GPU. We employed the AdamW optimizer with a batch size of 8 and implemented linear learning rate decay. The initial learning rate was set to 0.0001, while the momentum and weight decay were set to 0.99 and 0.01, respectively. The model was trained for a total of 200 epochs. To ensure fairness and enhance the model's robustness, data augmentation techniques such as flipping, cropping, and blurring were applied during the training phase for all experiments.

2) Evaluation Metrics: To quantitatively assess the effectiveness of CD methods, we employed five widely used metrics to compare the predicted change maps with the ground truth, including precision (Pre), recall (Rec), F1-score (F1), intersection over union (IoU), and overall accuracy (OA). Their definitions are as follows:

$$\mathrm{Pre} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \tag{16}$$

$$\mathrm{Rec} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \tag{17}$$

$$\mathrm{F1} = 2\times\mathrm{Rec}\times\mathrm{Pre}/(\mathrm{Rec}+\mathrm{Pre}) \tag{18}$$

$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}) \tag{19}$$

$$\mathrm{OA} = (\mathrm{TP}+\mathrm{TN})/(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}) \tag{20}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. They range from 0 to 1, where higher values indicate better performance.

Additionally, we also take into consideration the efficiency of the networks, including the number of parameters (Params) and the computation costs in terms of floating-point operations (FLOPs).

C. Comparison With SOTA Methods

The proposed SEIFNet was compared with eight representative methods, including fully convolutional early fusion (FC-EF) [21], IFN [34], SNUNet [25], bitemporal image transformer (BIT) [17], ChangeFormer [40], DMINet [36], TFI-GR [41], and A2Net [45]. FC-EF and SNUNet are fully convolutional networks with skip connections. The IFN incorporates attention mechanisms in feature fusion. BIT and ChangFormer are transformer-based models, where BIT combines CNN and transformer structures, while ChangFormer is a pure transformer architecture designed specifically for CD. Both TFI-GR and DMINet adopted feature interaction to capture temporal difference information at diverse feature levels. A2Net is a novel lightweight network with progressive feature aggregation and supervised attention. These approaches represent different ways of addressing the challenges of the CD task and share a similar starting point with the proposed SEIFNet. It is important to note that all methods were retrained with the same experimental conditions, following the official codes provided by the authors.

1) Quantitative Evaluation: The quantitative evaluations of various methods on LEVIR-CD, SYSU-CD, and WHU-CD datasets are presented in Table I. The evaluation metrics include OA, IoU, F1, Pre, and Rec. The quantitative evaluation results from the three datasets demonstrate the considerable superiority of the proposed SEIFNet. On the LEVIR-CD dataset, compared to the latest DMINet and A2Net, SEIFNet further improves IoU and recall to 83.40% and 89.46%, respectively. A reasonable explanation is that SEIFNet places more emphasis on spatial location differences and is better able to identify features that are ignored by other methods. In terms of quantitative results, A2Net achieved the highest precision, which proves the effectiveness of the progressive feature fusion; however, its recall needs to be improved, which may be related to the lightweight feature extraction network. In contrast to IFNet and SNUNet with large number of parameters, the lightweight structure of SEIFNet does not affect its performance. On the SYSU-CD dataset, SEIFNet also achieves the highest recall (84.81%) and precision (79.98%),

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TABLE I
QUANTITATIVE COMPARISONS ON CD RESULTS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods		FC-EF [21]	IFNet [34]	SNUNet [25]	BIT [17]	ChangeFormer [40]	DMINet [36]	TFI-GR [41]	A2Net [45]	SEIFNet (ours)
FLOPs (G)		1.35	82.26	54.83	15.17	21.18	14.55	9.74	3.05	8.37
Params (M)		3.57	50.44	12.03	17.99	29.75	6.24	27.78	3.78	27.91
LEVIR-CD (%)	OA	98.21	98.80	98.96	98.98	98.92	98.98	98.99	98.95	**99.09**
	IoU	69.46	77.91	81.36	81.52	80.60	81.64	82.95	80.58	**83.40**
	F1	81.98	87.58	89.72	89.82	89.26	89.90	90.68	89.24	**90.95**
	Pre	84.18	92.00	90.71	91.39	90.63	90.85	92.49	**92.96**	92.49
	Rec	79.89	83.58	88.75	88.30	87.92	88.96	88.94	85.81	**89.46**
SYSU-CD (%)	OA	89.18	89.41	90.45	89.57	89.68	91.37	91.47	91.63	**91.90**
	IoU	59.36	60.76	66.19	62.84	62.90	68.11	67.79	68.93	**69.96**
	F1	74.49	75.59	79.65	77.18	77.23	81.19	80.81	81.61	**82.32**
	Pre	83.91	82.78	80.04	79.73	80.47	84.19	**86.09**	84.69	84.81
	Rec	66.98	69.55	79.27	74.79	74.23	78.10	76.14	78.75	**79.98**
WHU-CD (%)	OA	97.35	98.88	98.68	98.80	98.50	98.64	98.46	98.89	**98.90**
	IoU	51.07	73.47	71.71	73.38	67.64	70.51	69.81	74.81	**76.04**
	F1	67.61	84.71	83.52	84.64	80.69	82.70	82.22	85.60	**86.39**
	Pre	67.09	95.72	84.62	88.07	84.37	85.12	77.40	**90.70**	87.01
	Rec	68.15	75.97	82.46	81.47	77.33	80.43	**87.69**	81.03	85.77

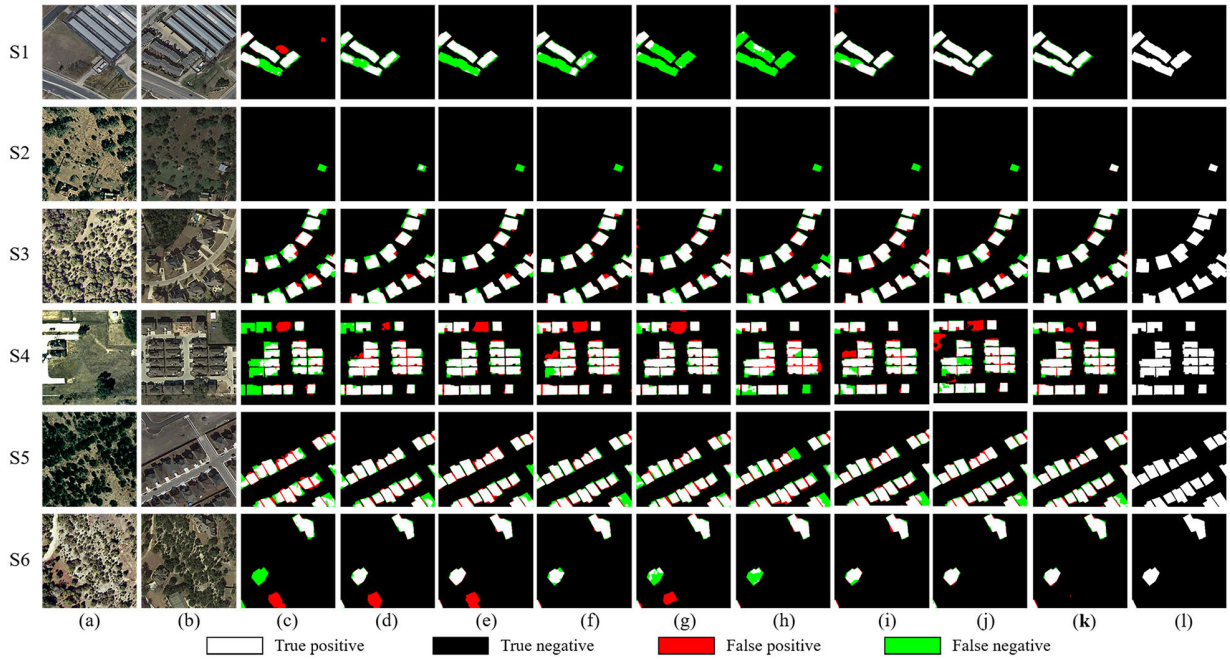


Fig. 4. Visual comparisons of the proposed method and SOTA methods on the LEVIR-CD dataset. (a) T_1 images. (b) T_2 images. (c) FC-EF [21]. (d) IFNet [34]. (e) SNUNet [25]. (f) BIT [17]. (g) ChangeFormer [40]. (h) DMINet [36]. (i) TFI-GR [41]. (j) A2Net [45]. (k) Proposed SEIFNet. (l) Ground truth. S1–S6 represent six representative samples.

which demonstrates that our method excels better on more challenging datasets. From the perspective of efficiency, the computational time of SEIFNet is one-half that of MDNIet, while F1 and IOU are both improved by more than 2%. On the WHU-CD dataset, SEIFNet achieved a noticeable advantage in the synthesized evaluation metrics, e.g., IoU and F1. A2Net and DMINet achieved the highest values on Pre and Rec. This indicates that the proposed method has a better balance between misdetection and omission, but there is still some room for improvement. Quantitative evaluation results demonstrate the effectiveness and superiority of the SEIFNet method.

2) Qualitative Evaluation: The visual comparisons of different methods in two datasets are shown in Figs. 4–6, where distinct colors are assigned to identify the correctness or incorrectness of the results, including true positives (white), false positives (red), true negatives (black) and false negatives (green). From the results of some challenging samples, the proposed method is superior in the following aspects.

a) Advantages in reducing pseudo changes: For the building CD task, reducing the interference of pseudo changes and shadows are two difficult problems. As shown in Fig. 4 (S4), Only SEIFNet accurately detected all buildings and clearly predicted their boundaries. Comparing the false and missed detections in Figs. 5 and 6, SEIFNet shows remarkably accurate discriminability even under occlusion and illumination interference. These observations demonstrate the

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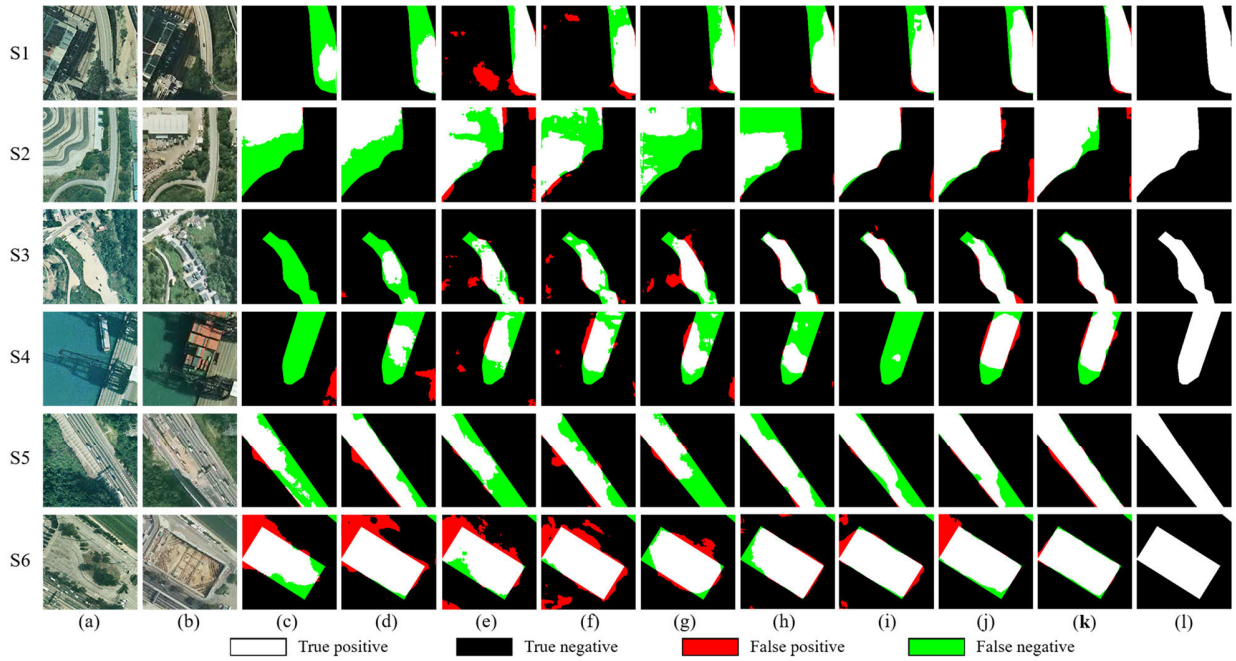


Fig. 5. Visual comparisons of the proposed method and SOTA methods on the SYSU-CD dataset. (a) T_1 images. (b) T_2 images. (c) FC-EF [21]. (d) IFNet [34]. (e) SNUNet [25]. (f) BIT [17]. (g) ChangeFormer [40]. (h) DMINet [36]. (i) TFI-GR [41]. (j) A2Net [45]. (k) Proposed SEIFNet. (l) Ground truth. S1–S6 represent six representative samples.

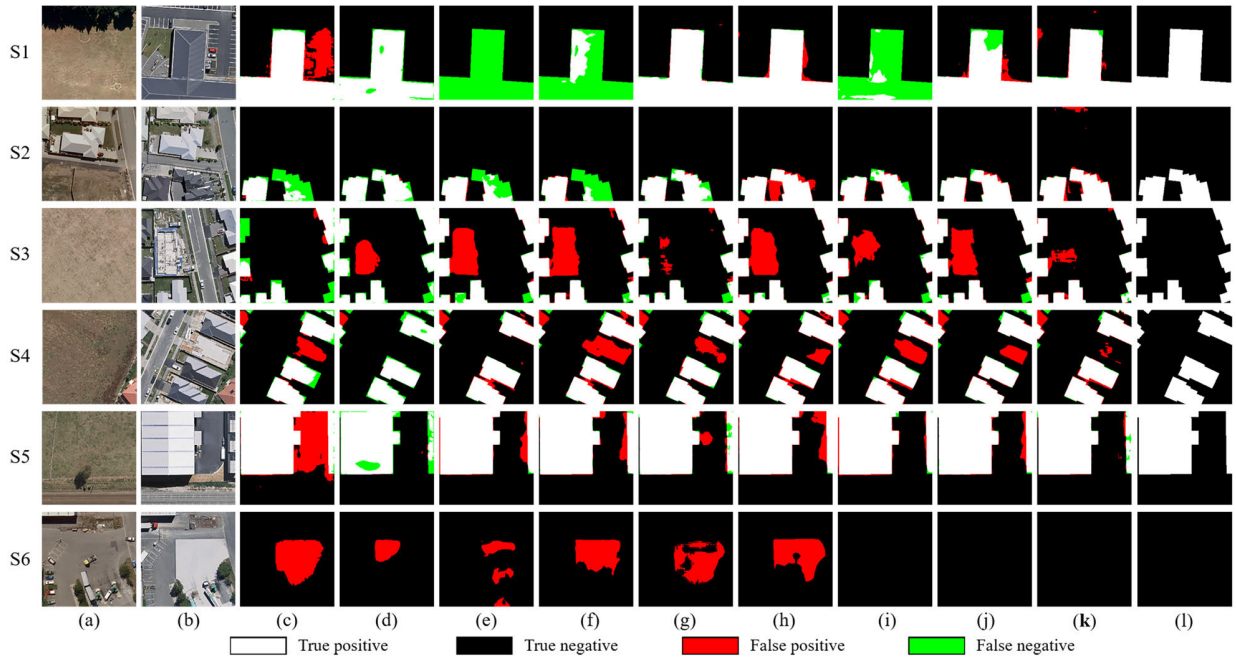


Fig. 6. Visual comparisons of the proposed method and SOTA methods on the WHU-CD dataset. (a) T_1 images. (b) T_2 images. (c) FC-EF [21]. (d) IFNet [34]. (e) SNUNet [25]. (f) BIT [17]. (g) ChangeFormer [40]. (h) DMINet [36]. (i) TFI-GR [41]. (j) A2Net [45]. (k) Proposed SEIFNet. (l) Ground truth. S1–S6 represent six representative samples.

effectiveness of ST-DEM in highlighting relevant changes and suppressing irrelevant interference.

b) Advantages in distinguishing scale variations: Large objects are susceptible to incomplete detection due to receptive field limitations, while features of small objects are prone to be lost. For example, as shown in Fig. 4 (S1), large complexes that are close to the background are prone to being missed by other methods; however, SEIFNet [Fig. 4(j)] successfully detects them. SEIFNet can accurately detect such small buildings that undergo changes in Fig. 4 (S2) as well. It demonstrates the superiority of SEIFNet in capturing and aggregating multilevel differences. This indicates that multiscale feature fusion facilitates information complementarity and thus reduces missing features.

c) Advantages in edges and details: There are mutual constraints between the accuracy of edge detection and the completeness of internal details. In Fig. 5 (S6) and Fig. 6 (S5), several methods struggle to identify the difference between similar-looking objects such as roads and bare ground. Among them, tolerable CD results are achieved by MDINet and

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TABLE II
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS ON THREE DATASETS

No.	Variants	Efficiency		LEVIR-CD (%)	SYSU-CD (%)	WHU-CD (%)
		FLOPs (G)	Params (M)	IoU / F1/ Rec	IoU / F1/ Rec	IoU / F1 / Rec
#01	SEIFNet	8.37	27.91	83.40 / 90.95 / 89.46	69.96 / 82.32 /79.98	76.04 / 86.39 / 85.77
Different Backbones						
#02	MobileNetV2	1.54	7.64	82.33 / 90.31 / 88.83	68.11 / 81.03 / 78.90	73.53 / 84.75 / 84.95
#03	Transformer	13.04	47.46	80.79 / 89.33 / 87.62	65.93 / 79.47 / 76.26	72.25 / 83.89 / 82.10
ST-DEM						
#04	w/o ST-DEM	5.81	14.60	80.61 / 89.27 /87.83	67.75 / 80.78 / 77.12	71.74 / 83.55 / 84.81
#05	FDEM	5.88	15.30	80.34 / 89.10 / 86.41	68.58 / 81.36 / 78.35	72.08 / 83.77 / 85.55
#06	DM	7.63	24.00	81.76 / 89.97 / 87.79	68.05 / 80.99 / 77.86	65.91 / 79.46 / 85.88
#07	DEFM	5.97	14.98	80.38 / 89.13 / 86.59	66.89 / 80.16 / 77.40	71.77 / 83.57 / 84.26
ACFM						
#08	w/o ACFM	8.37	27.89	81.81 / 90.00 /88.86	69.09 / 81.72 /78.66	74.01 / 85.07 / 84.68
RM						
#09	w/o RM	7.61	24.69	82.43 / 90.76 / 88.67	67.76 / 81.07 / 75.62	74.69 / 85.84 / 85.26

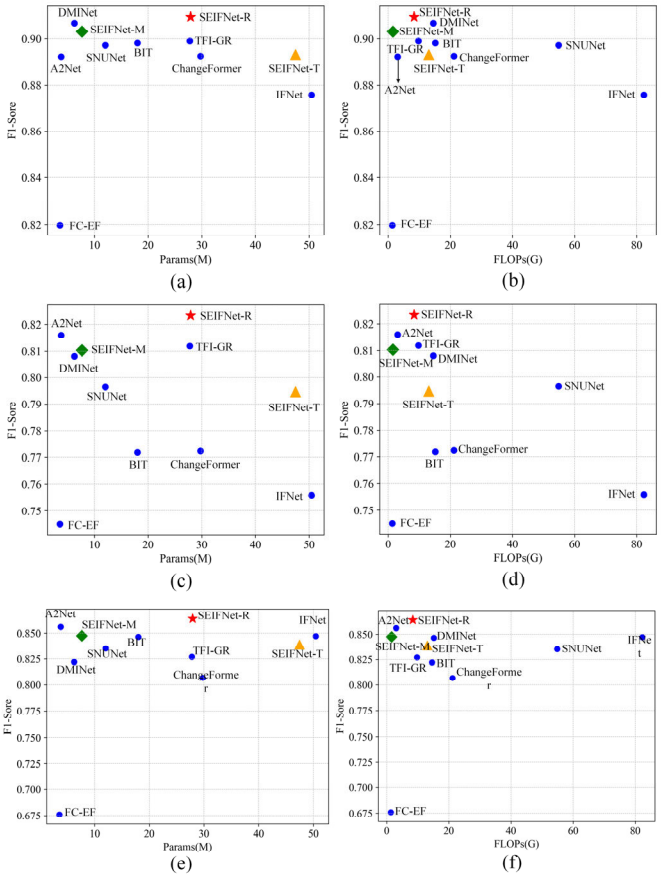


Fig. 7. Model complexity comparison of different backbones and methods on the LEVIR-CD, SYSU-CD and WHU-CD datasets. (a) and (b) LEVIR-CD dataset. (c) and (d) SYSU-CD dataset, (e) and (f) WHU-CD dataset. The backbone of SEIFNet-R, SEIFNet-M, and SEIFNet-T are ResNet18, MobileNetv2, and hierarchical transformer encoder, respectively.

A2Net, but changes in the edge regions are prone to being missed. In contrast, SEIFNet effectively identifies the boundaries and bodies of objects. These results can be attributed to the progressive decoder with context-guided features and the multilevel refined features aggregation, which enhance content integrity and improve accuracy.

Hence, SEIFNet demonstrates remarkable visualization results that align closely with Table I, consistently achieving higher true positive rates and lower false positive rates.

V. DISCUSSION

In order to validate the effectiveness of the components and configurations of SEIFNet, the ablation experiments are conducted on three datasets. Besides, its current limitations are also analyzed in this section.

A. Effects of Different Backbones

To illustrate the rationality of the proposed model, different backbones are used to further validate the availability, including MobileNetV2 [51] and hierarchical transformer encoder [40], which are commonly used for feature extraction in CD alongside ResNet18. As seen from Table II, SEIFNet with ResNet18 (#01) achieves superior performance compared to using MobileNetV2 (#02) and hierarchical transformer encoder (#03) on both LEVIR-CD, SYSU-CD and WHU-CD datasets.

To provide a clearer demonstration of the impact of various backbones in our proposed method, we show the accuracy in terms of F1-score, Params, and FLOPs of SEIFNet with ResNet18, MobileNetV2, and hierarchical transformer encoder, along with the comparative methods mentioned in Section IV-C. The three kinds of variants are denoted as SEIFNet-R, SEIFNet-M, and SEIFNet-T in Fig. 7, respectively. As seen from the results, the highest accuracy on three datasets was achieved by SEIFNet-R, with a reasonable model complexity; however, it is worth noting that SEIFNet-M demonstrates remarkable performance in terms of both efficiency and accuracy. In comparison to SEIFNet-R, SEIFNet-M exhibits a significant reduction in FLOPs and Params by 80.6% and 72.6%, respectively, while only experiencing a slight decrease of 0.64% in F1-score. These findings indicate that the designed structure is competitive in terms of lightweight models. Although SEIFNet-T may not achieve the same level of accuracy as the previous two variants, it outperforms

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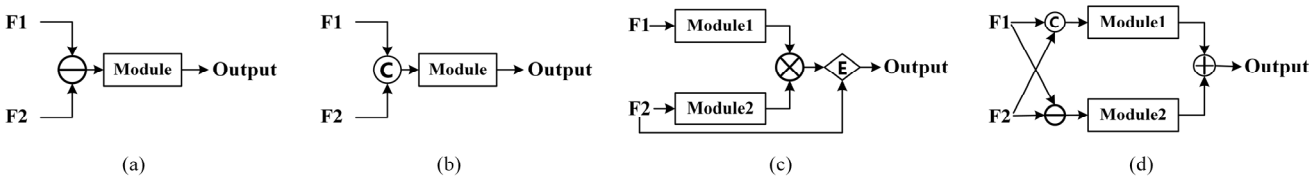


Fig. 8. Simplified schematic of different difference enhancement modules and proposed ST-DEM. (a) Difference-based enhancement (e.g., FDEM [52]). (b) Connection-based enhancement (e.g., DM [40]). (c) Single image enhancement (e.g., DEFM [54]). (d) Proposed ST-DEM.

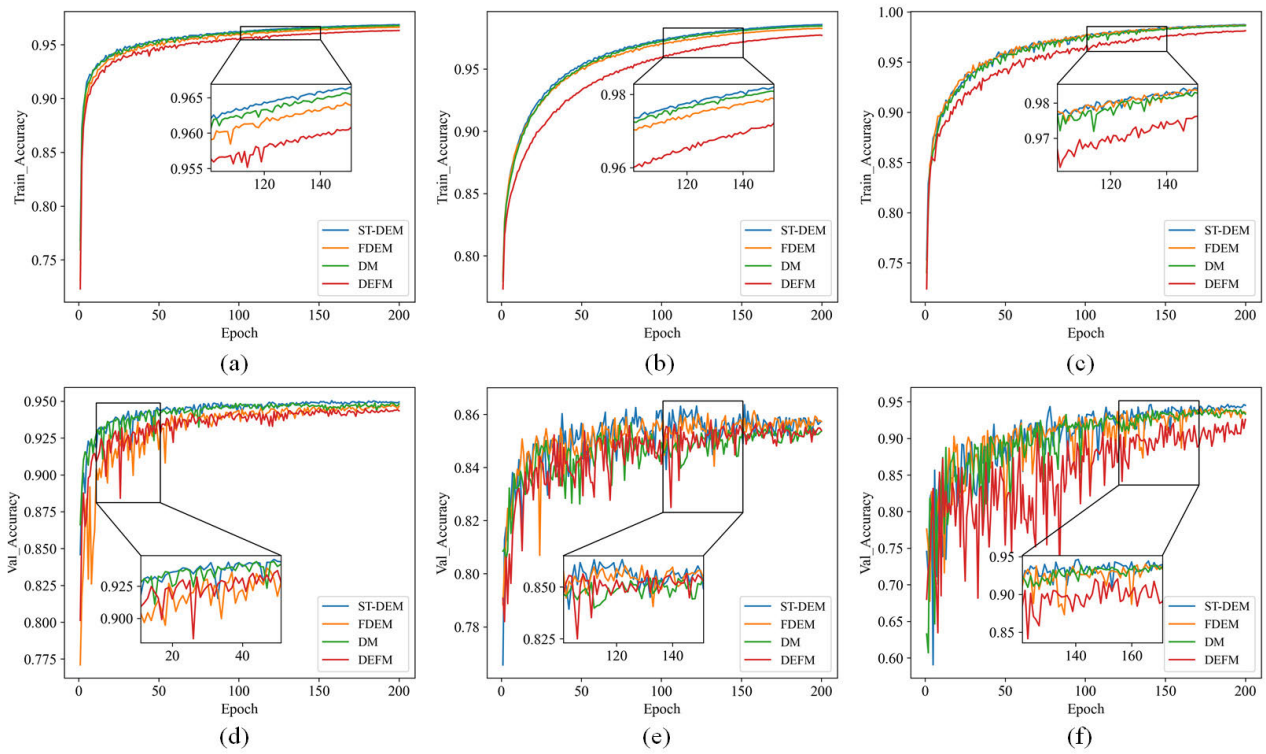


Fig. 9. Training and validation accuracy variations against the increasing training epoch of different difference enhancement modules and proposed ST-DEM. (a) and (d) On the LEVIR-CD dataset. (b) and (e) On the SYSU-CD dataset, (c) and (f) On the WHU-CD dataset.

ChangeFormer with the same backbone. In summary, the proposed method with ResNet18 achieves the highest accuracy among the tested backbones. If, however, there is a need for lightweight models, MobileNetV2 is a suitable choice due to its efficiency. From another point of view, our proposed model is highly flexible and can benefit from future advancements in feature extraction methods, allowing for continuous improvement.

B. Effects of Different Modules

1) Effectiveness of ST-DEM: The proposed SEIFNet incorporates the ST-DEM module to improve the ability to capture difference maps between bitemporal feature maps.

To evaluate the effectiveness of the ST-DEM module, an experiment was conducted by replacing it with a simple absolute subtraction operation. The variant is denoted as w/o ST-DEM (#04 in Table II), and the corresponding quantitative results are reported. The results show that when ST-DEM is replaced with a simple absolute subtraction, the detection accuracy in terms of IoU and F1 decreases by 2.79% and 1.68% on the LEVIR-CD dataset, decreases by 2.21% and 1.54% on the SYSU-CD dataset, and 4.3% and 2.84% on WHU-CD, respectively. The results highlight the significance of ST-DEM, as it plays a crucial role in improving the performance.

To further illustrate the advantages of ST-DEM, it was compared with the other three kinds of difference enhancement modules commonly used in RSCD. These modules aim to enhance the differences in bitemporal feature maps for better CD results; however, their implementations are totally different. As shown in Fig. 8, the first type is difference-based enhancement [Fig. 8(a)], such as the feature differential enhancement module (FDEM) [53]. In FDEM, the differential map is obtained by subtracting the two feature maps. The second one is the connection-based enhancement approach [Fig. 8(b)], like difference module (DM) [40], which learns the optimal distance metric by concatenating bitemporal feature maps and applying a 3×3 convolutional operation. Another common method is achieved with single image enhancement, as shown in Fig. 8(c). One typical method is a difference enhancement fusion module (DEFM) [54].

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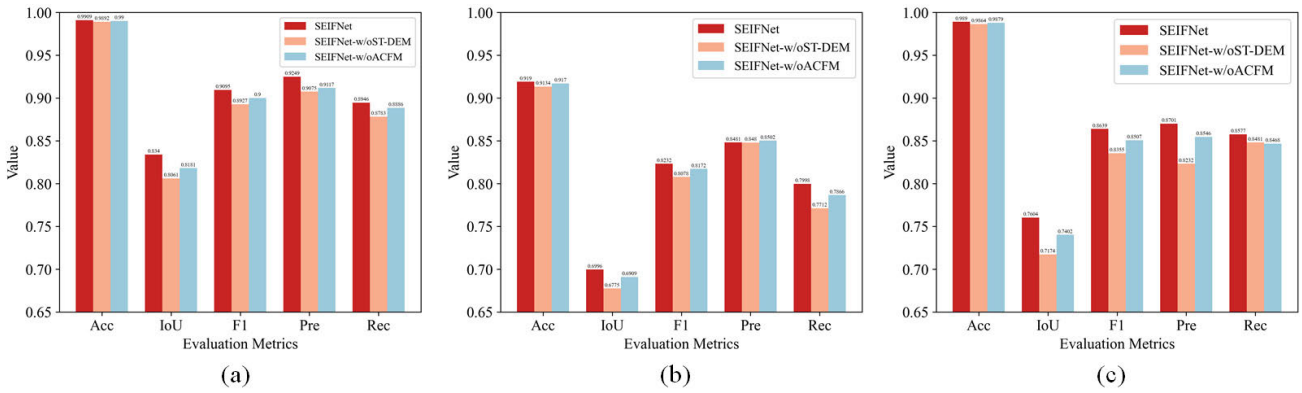


Fig. 10. Visualization of the effectiveness of ST-DEM and ACFM. (a) LEVIR-CD dataset. (b) SYSU-CD dataset. (c) WHU-CD dataset.

TABLE III
QUANTITATIVE COMPARISONS OF DIFFERENT COEFFICIENTS OF LOSS FUNCTION ON THREE DATASETS

λ	LEVIR-CD (%) IoU / F1 / Rec	SYSU-CD (%) IoU / F1 / Rec	WHU-CD (%) IoU / F1 / Rec
0.00	81.58 / 89.86 / 88.00	68.13 / 81.04 / 76.56	73.69 / 84.15 / 82.24
0.25	81.90 / 90.05 / 88.29	68.00 / 80.95 / 76.11	74.58 / 85.23 / 83. 46
0.50	81.58 / 89.86 / 88.15	68.64 / 81.41 / 78.38	75.22 / 85.47 / 83.94
0.75	**83.40 / 90.95 / 89.46**	68.77 / 81.50 / 77.58	**76.04 / 86.39 / 85.77**
1.00	82.12 / 90.18 / 88.13	**69.96 / 82.32 / 79.98**	75.86 / 85.68 / 84.21

DEFM expands the difference between change-related and change-irrelevant regions within a single feature map by the learned time-position offset of corresponding pixels in bitemporal feature maps. The results in Table II (#05–#07) demonstrate that the ST-DEM achieves the best performance, indicating the effectiveness of learning global and local context information with a dual branch; furthermore, Fig. 9 illustrated the increased training and validation accuracy against the epochs on three datasets. Among them, the training accuracy of ST-DEM is always the highest. Fig. 9(d), (e), and (f) demonstrate that ST-DEM exhibits a more stable curve on the validation set compared to the other modules. This confirms the strong robustness and superiority of our proposed ST-DEM module.

2) Effectiveness of ACFM and RM: The ACFM module is designed to capture fused features by selecting contextual information from difference-enhanced features in neighboring stages. RM is introduced to optimize edge details during decoding. To evaluate the effectiveness of ACFM, an ablation experiment was conducted by removing the weight calculation part of adaptive fusion. Instead, the addition of upsampled high-level features and low-level features is used as the fusion result, referred to as w/o ACFM (#08 in Table II). The experimental results in Table II clearly demonstrate the superiority of SEIFNet over SEIFNet-w/o ACFM, indicating the importance of adaptive weight learning in our proposed method. Experiments are conducted by removing rms to verify its impact on network performance, referred to as w/o RM (#09 in Table II). As can be seen from Table II, the designed RM can further improve the accuracy of the CD. It is attributed to the fact that the boundary information is preserved by paying attention to the detailed features during the decoding process. The impact of removing ST-DEM and ACFM in SEIFNet is visualized in Fig. 10. ACFM may not show a significant improvement in F1-score compared to ST-DEM, but it plays a crucial role in optimizing the details of the results. It further highlights the significance of incorporating both high-level and low-level features. It helps in decoding to maintain feature integrity, and thus, improves the overall performance of the network.

3) Coefficients of Class Balance Loss Function: To assess the impact of the class balance loss function, we conducted a study by varying the tradeoff coefficients for the BCE loss and Dice loss. To maintain a balance between the two loss functions during training, λ was selected from the range of $[0, 1]$ with an interval of 0.25. The results are presented in Table III, which demonstrates that our proposed method is robust with different parameters. It is, however, observed that the optimal parameter settings differ for different datasets. Specifically, on the LEVIR-CD and WHU-CD datasets, the best performance was achieved with a value of λ equal to 0.75, while on the SYSU-CD dataset, the results were even better when λ was set to 1. Indeed, the observed variation in performance can be attributed to the significant disparities in sample distributions between the two datasets. To simplify model selection, we recommend setting $\lambda = 0.75$ as a practical choice.

4) Visualization of weights in ACFM: In Fig. 11, samples from the three datasets were selected to demonstrate the distribution of weights of Fusion[3] in the ACFM. The differences between weighted (#1, #3 in Fig. 11) and unweighted (#2, #4 in Fig. 11) fusion strategies are compared by activation maps and heatmaps. The brightness in the activation map indicates

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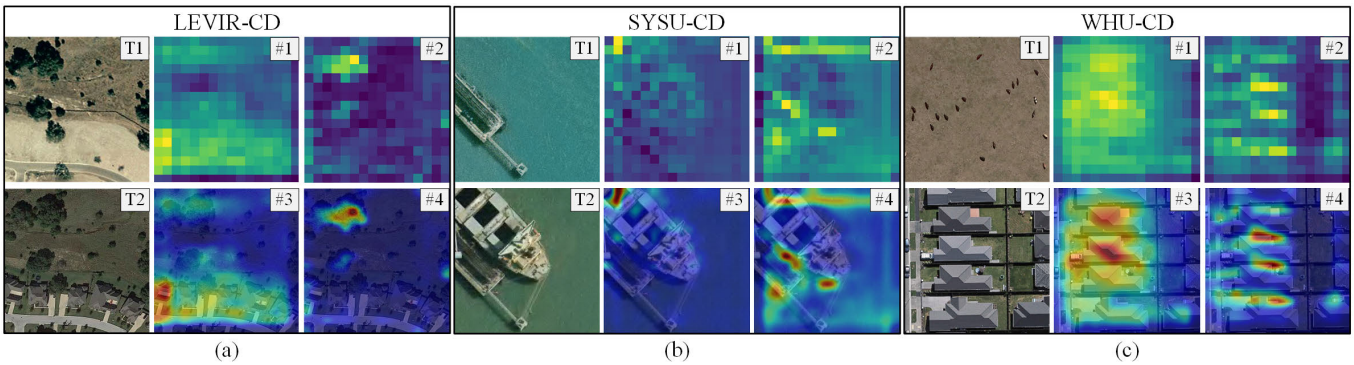


Fig. 11. Visualization of weights in ACFM. (a) LVEIR-CD dataset. (b) SYSU-CD dataset. (c) WHU-CD dataset. T1 and T2 represent the bitemporal inputs. #1 and #2 are activation maps of ACFM with and without weight fusion, respectively. #3 and #4 are heatmaps of ACFM with and without weight fusion, respectively.

the rank of the weights; the brighter it is, the higher the weights are. The heatmaps based on the Grad_CAM [55] visualize the spatial location of attention. From the visualization results, the weights obtained from adaptive learning enhance the feature representation. At the same time, the spatial location of changing features can also be accurately localized, as shown in Fig. 11(a) and (b). This proves that the introduction of weighted fusion integrates the advantages that are contained in both deep and shallow features. The design of ACFM enables the preservation of feature integrity and spatial details during the decoding process.

C. Limitations

During the experimental process, we also found some shortcomings in the present methodology. From the CD results, although our method significantly reduces misdetections (false positive), it can be prone to omission at the image edge. For example, the false negative is present in the lower right corner of Fig. 4 (S5) and the right side of Fig. 6 (S5). We analyze that the shallow convolutional operation may easily cause the loss of features at the edges, which can no longer be acquired in the subsequent learning. This also indicates that the coding part of the model is crucial. In the follow up, we will continue to explore the construction of feature extraction models adapted to the distribution of objects in RS images. Apart from that, the model with MobileNet2 as the backbone still has optimization space in the detection results. How to balance efficiency and performance is also the target direction of future research.

VI. Conclusion

To investigate the contextual information of changes between bitemporal RS images, SEIFNet is proposed with spatiotemporal difference enhancement and adaptive context fusion for the CD task. Initially, hierarchical features are extracted from the bitemporal images using a Siamese pretrained ResNet18 as the backbone. To focus on genuine changes and suppress irrelevant changes, ST-DEM is introduced to capture global spatiotemporal and contextual information between the two feature maps. Subsequently, the multiscale difference features are decoded by ACFM and refinement module, which adaptively reweight the compact internal structure and position details. Finally, the refined features at different scales are combined to achieve more comprehensive and precise CD results. Extensive experiments on three publicly available datasets demonstrate the superiority of our proposed SEIFNet over eight SOTA methods in terms of both efficiency and effectiveness. Considering the flexibility of our model in feature extraction, future research aims to explore a more powerful backbone that is suitable for interpreting bitemporal RS images.

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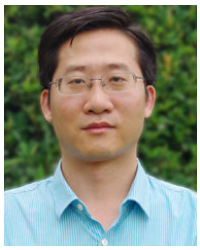
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